FashionFusion

Project Team

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1 Introduction

1.1 Problem Statement

When shopping online, customers frequently face the challenge of browsing through multiple websites to explore products from different brands. This process is cumbersome, especially when comparing similar products across different platforms. Many brands do not offer comparison features, making it difficult to evaluate products side-by-side. Moreover, a significant number of e-commerce platforms in Pakistan lack review systems, which results in a lack of credibility and transparency.

Additionally, customers are often hesitant to purchase products online due to the uncertainty of store authenticity and product quality. The lack of performance evaluation and customer feedback on many local platforms further aggravates the issue. **FashionFusion** aims to solve these problems by aggregating products from multiple clothing brands, enabling product comparison, and integrating a user review system. The platform will also provide brand performance scores to build trust among customers and enhance the online shopping experience.

1.2 Scope

Products from different brands will be scraped and stored in our database. A responsive website including major functionalities of an ecommerce platform (like searching and filtering) will be built and the products stored in the database will be displayed on the website. This website will also include product comparison features and a reviews system. After creating the basic website the scraper will be set to fetch products fortnightly to keep the platform updated. Campaigns will be monitored and sentiment analysis will also be done to analyze brands' performance and create a brand's picture. For campaign monitoring prices will be regularly checked for change whether a brand provides a year long sale or it provides occasional sale and do the brands hike prices before sales. For sentiment analysis dataset will be gathered from scraping the brands' social media comments, this data will then be used to train the model. The model will then be integrated in the website. A pre-trained LLM Chatbot will then be integrated to enhance the customer's shopping experience.

1.3 Modules

The Modules of this Project are as follows:

1. Web Scraping

To start with this project, the first requirement is to gather products from different brands' websites and store them on a unified database. For sentiment Analysis comments and reviews from brands' social media pages will be scraped to create a dataset.

1. Products Scraping
2. Comments and Reviews Scraping

2. Frontend Module

This is the Users' interaction interface where they can access all the functionalities of Fashion Fusion.

1. Responsive UI / UX
2. Advanced Searching and Filters
3. Comparison Feature
4. Reviews Feature

3. Backend Module

The brains and nervous system of the FashionFusion is its secure backend which will be responding to the user's request sent through the frontend.

1. Authentication
2. RestAPI
3. Database

4. Sentiment Analysis Module

From the scraped data (reviews and comments), data will be refined [33] and a sentiment analysis model will be trained [22] which will then be integrated in the website. This model will be kept as self training [11] so that it can update brands' score

1. Data cleaning
2. Training the Model
3. Integrating it with the Website
4. Continuous Model Training

5. LLM

User will be able to communicate with a chatbot who will give information to the user regarding a certain brand. It will also help users to easily search for products and take recommendations

6. Report Generation Module

On the basis of campaigns/price monitoring and sentiment analysis brands scoring will be created and the brands will be ranked accordingly.

1. Brand Scoring and Ranking
2. Product Scoring

1.4 User classes and Characteristics

User Class	Description
Customers	Regular customers who shop for clothes across different brands. They seek convenience in comparing products and value user reviews.
Admin Team	Team members responsible for maintaining the platform, managing brand data, and overseeing user interactions. They require access to all functionalities of the platform.

2 Project Requirements

2.1 Use Cases

Use Case 1: Product Sraping

- **Actors: Web Scraper**
- **Description: The system scrapes product data from various brand websites and stores it in the database.**
- **Preconditions: The web scraper is initialized, and the target websites are accessible.**
- **Postconditions: New product data is successfully added to the database.**

Use Case 2: Comment and Review Scraping

- **Actors: Web Scraper**
- **Description: The system gathers comments and reviews from social media pages of brands to create a dataset for analysis.**
- **Preconditions: The web scraper is set up to target specific social media platforms.**
- **Postconditions: Collected comments and reviews are stored in the database for future analysis.**

Use Case 3: User Registration

- **Actors: User**
- **Description: A user registers for an account on the FashionFusion platform.**
- **Preconditions: The user has access to the internet and a valid email address.**
- **Postconditions: A new user account is created and a confirmation email is sent.**

Use Case 4: User Authentication

- **Actors: User**
- **Description: A user logs into their FashionFusion account using their credentials.**
- **Preconditions: The user must have a registered account.**
- **Postconditions: The user is authenticated and granted access to the platform.**

Use Case 5: Search Products

- **Actors: Customer**
- **Description: A user searches for products using keywords or filters.**
- **Preconditions: The user is logged into the platform.**
- **Postconditions: The system displays a list of products that match the search criteria.**

Use Case 6: Compare Products

- **Actors: Customer**
- **Description: A user compares multiple products side by side.**
- **Preconditions: The user has selected products for comparison.**

- **Postconditions: The system displays a comparison of the selected products, highlighting differences and similarities.**

Use Case 7: Give Review

- **Actors: Customer**
- **Description: A user submits a review for a product they have purchased.**
- **Preconditions: The user has purchased the product and is logged into the platform.**
- **Postconditions: The submitted review is stored in the database and associated with the product.**

Use Case 8: Review Analysis

- **Actors: System**
- **Description: The system analyzes collected comments and reviews to generate sentiment scores for brands.**
- **Preconditions: There are enough reviews and comments in the database for analysis.**
- **Postconditions: Sentiment scores are updated and available for users to view.**

Use Case 9: Rank Brands

- **Actors: System**
- **Description: The system generates scores for brands based on sentiment analysis and performance metrics.**
- **Preconditions: The sentiment analysis model has been trained and data is available.**
- **Postconditions: Brand scores are calculated and stored in the database for retrieval by users.**

Use Case 10: Chatbot Interaction

- **Actors: Customer**
- **Description: A user interacts with the chatbot to receive information and recommendations regarding products and brands.**
- **Preconditions: The user is logged into the platform.**
- **Postconditions: The chatbot provides relevant information based on user queries.**

Use Case 11: Advanced Filtering

- **Actors: Customer**
- **Description: A user applies filters to narrow down product search results based on specific criteria (e.g., price, category, brand).**

- **Preconditions: The user is on the product search page.**
- **Postconditions: The system displays filtered results based on user selections.**

Use Case 12: Manage Profile

- **Actors: Customer , Admin**
- **Description: A user updates their profile information, including preferences and saved searches.**
- **Preconditions: The user is logged into their account.**
- **Postconditions: The user's profile is updated with new information.**

Use Case 13: Report Generation

- **Actors: System**
- **Description: Report is generated based on brand scores and product performance.**
- **Preconditions: The user has access to the report generation feature.**
- **Postconditions: The system produces a downloadable report summarizing the requested data.**

Use Case 14: Continuous Model Training

- **Actors: System**
- **Description: The system periodically retrains the sentiment analysis model with new data to improve accuracy.**
- **Preconditions: New data is available for training.**
- **Postconditions: The sentiment analysis model is updated with enhanced capabilities.**

Use Case 15: Real-Time Price Monitoring

- **Actors: System**
- **Description: The system monitors product prices in real time and alerts users to changes.**
- **Preconditions: The user has opted in for price alerts.**
- **Postconditions: The user receives notifications about price changes.**

Use Case 16: View Activity Log

- **Actors: Customer**
- **Description: A user views their activity log, including recent searches and interactions.**
- **Preconditions: The user is logged into their account.**
- **Postconditions: The user can see a summary of their activities on the platform.**

2.2 Event Response Table

Event	System Response
Web Scraper initiates product scraping	The system scrapes product data from target websites and stores it in the database.
Web Scraper initiates comment and review scraping	The system gathers comments and reviews from social media and stores them in the database.
User submits registration form	The system creates a new user account and sends a confirmation email.
User submits login credentials	The system validates credentials and grants access if correct; otherwise, displays an error message.
User performs a product search	The system displays a list of products matching the search criteria.
User selects products for comparison	The system displays a comparison of the selected products, highlighting differences and similarities.
User submits a product review	The system stores the review in the database and associates it with the product.
Sentiment analysis is triggered	The system analyzes comments and reviews to generate sentiment scores for brands.
Brand scoring is triggered	The system generates scores for brands based on sentiment analysis and performance metrics.
User interacts with the chatbot	The chatbot responds with relevant information and recommendations based on user queries.
User applies filters to search results	The system displays filtered product results based on user selections.
User updates profile information	The system updates the user’s profile with new information.
User requests a report	The system processes the request and produces a downloadable report summarizing brand scores and product performance.

Continuous model training is initiated	The system retrains the sentiment analysis model with new data to improve accuracy.
User submits feedback	The system records the feedback for future improvements to the platform.
User compares products	The system shows a comparison table of selected products.
User requests real-time price monitoring	The system monitors product prices and alerts users to any changes.
User views activity log	The system displays a summary of the user's activities on the platform.

2.3 Functional Requirements

Module 1: Web Scraping

Following are the requirements for the Web Scraping module:

1. The system shall scrape product information from various clothing brand websites, including details such as product name, price, description, and image URL.
2. The system shall scrape comments and reviews from the brands' social media pages to create a dataset for sentiment analysis.
3. The system shall store the scraped product data and reviews in a unified database for easy access and retrieval.

Module 2: Frontend Module

Following are the requirements for the Frontend module:

1. The system shall provide a responsive UI/UX to enhance user interaction and accessibility across devices.
2. The system shall offer advanced searching and filtering options for users to find products easily based on various criteria (e.g., price, brand, category).
3. The system shall allow users to compare multiple products from different brands on a single screen.
4. The system shall enable users to post and view reviews for products, enhancing user feedback and interaction.

Module 3: Backend Module

Following are the requirements for the Backend module:

1. The system shall implement user authentication to ensure secure access to the platform.
2. The system shall provide a REST API to facilitate communication between the frontend and backend components.
3. The system shall ensure a secure database to store user information, product details, and reviews.

Module 4: Sentiment Analysis Module

Following are the requirements for the Sentiment Analysis module:

1. The system shall perform data cleaning on scraped reviews and comments to prepare them for analysis.
2. The system shall train a sentiment analysis model using the cleaned data to assess customer sentiment towards brands.
3. The system shall integrate the trained sentiment analysis model with the website to provide real-time sentiment assessment.
4. The system shall implement continuous model training to keep the sentiment analysis updated based on new data.

Module 5: LLM (Chatbot)

Following are the requirements for the LLM module:

1. The system shall allow users to communicate with a chatbot that provides information about specific brands.
2. The system shall enable the chatbot to assist users in searching for products and providing recommendations based on user queries.

Module 6: Report Generation Module

Following are the requirements for the Report Generation module:

1. The system shall generate brand scoring based on price monitoring and sentiment analysis results.
2. The system shall rank brands accordingly based on their scores, allowing users to see the best-performing brands.
3. The system shall generate reports detailing product scoring and brand performance metrics.

2.4 Non-Functional Requirements

Reliability

The following show how reliable the system would be :

1. The FashionFusion system shall maintain a Mean Time Between Failures (MTBF) of at least 30 days, ensuring that software failures occur infrequently.
2. In the event of a software failure, the system shall log the failure details and notify the technical support team within 5 minutes of detection.
3. The system shall implement error detection mechanisms to identify anomalies and provide automatic alerts for system administrators.

Usability

The following show the usability of the system :

1. The system shall allow users to retrieve their previously viewed products with a single interaction, enhancing the user experience.
2. The system shall provide clear and concise feedback to users for any errors encountered, helping them recover from mistakes efficiently.
3. The user interface shall be designed to require no more than three clicks to reach any main feature, ensuring an efficient navigation experience.

Performance

The following steps would be taken to ensure best performance :

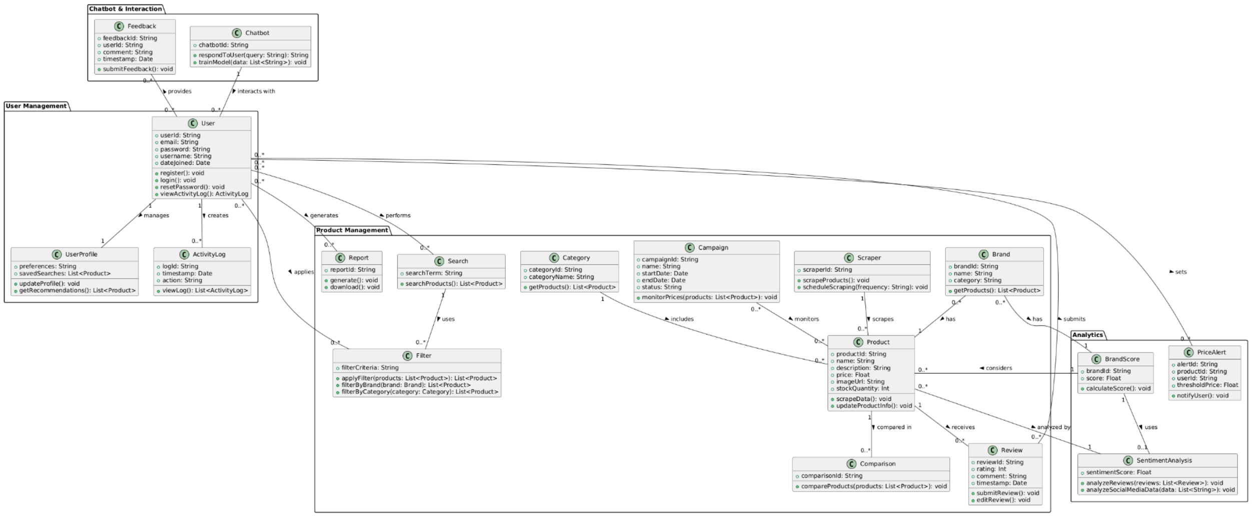
1. The system shall ensure that 95% of product pages load completely within 3 seconds over a standard broadband connection (at least 20 Mbps).
2. The sentiment analysis module shall process and update brand scores within 10 seconds after new reviews are scraped, maintaining up-to-date information for users.
3. The system shall handle at least 100 concurrent users without performance degradation, ensuring a seamless experience during peak usage.

Security

This is how security will be ensured :

1. The FashionFusion system shall implement measures to protect user data and prevent unauthorized access, ensuring that only authenticated users can access sensitive information.
2. The system shall encrypt all sensitive data in transit and at rest to safeguard against potential data breaches.
3. The system shall maintain a comprehensive logging system to monitor access and changes to user accounts and sensitive data, allowing for audit trails in case of security incidents.

2.5 Domain Model



3 System Overview

FashionFusion is a review-based platform designed to provide users with valuable insights into fashion brands and products. It collects data from various sources, performs sentiment analysis on user reviews, and presents the results in an interactive and user-friendly interface.

The platform allows users to search and filter products based on categories, brands, and other criteria, providing a comprehensive comparison and review system. It also monitors price changes and campaigns over time, giving users an in-depth view of brand performance.

With features like a chatbot for interaction, report generation, and sentiment analysis, FashionFusion aims to enhance users' understanding of the fashion market and support better decision-making.

3.1 Architectural Design

The architecture of the FashionFusion platform is based on a Layered Architecture Pattern, which divides the system into multiple layers, each responsible for specific types of functionalities. This layered approach facilitates maintainability, scalability, and modular development, ensuring that each module can be developed, tested, and updated independently.

Overview of Modules

The platform is divided into the following primary modules:

1. User Management Module:

    - Manages user-related operations such as registration, login, and profile management.
    - Provides user authentication and authorization to ensure secure access to system functionalities.
2. Product Management Module:
    - Handles operations related to product data, such as scraping new products and updating product prices.
    - Manages the storage and retrieval of product information from the database.
3. Review Management Module:
    - Manages all review-related functionalities, including submission, editing, and deletion of reviews.
    - Stores user reviews and links them to respective products.
4. Analytics Module:
    - Processes user reviews and feedback to generate sentiment scores and track trends.
    - Analyzes product data to monitor changes in product prices and detect sales patterns.
5. Chatbot Module:
    - Provides an interactive medium for users to ask questions and receive product recommendations or information.
    - Integrates data from other modules to provide real-time responses to user queries.
6. Report Generation Module:
    - Generates detailed reports based on analytics results and product information.
    - Produces visualizations and insights for users to make informed decisions.

Module Interactions and Relationships

The modules interact with each other as follows:

- The User Management Module provides authenticated access to the platform and collaborates with the Product Management and Review Management modules to store user activities, such as product views and review submissions.
- The Product Management Module and Review Management Module serve as data sources for the Analytics Module, which processes this data to generate insights such as sentiment scores and trends.
- The Analytics Module communicates with the Report Generation Module to produce comprehensive reports, providing users with an easy-to-understand summary of findings.
- The Chatbot Module pulls data from multiple modules to deliver accurate responses to user queries, ensuring seamless interaction.

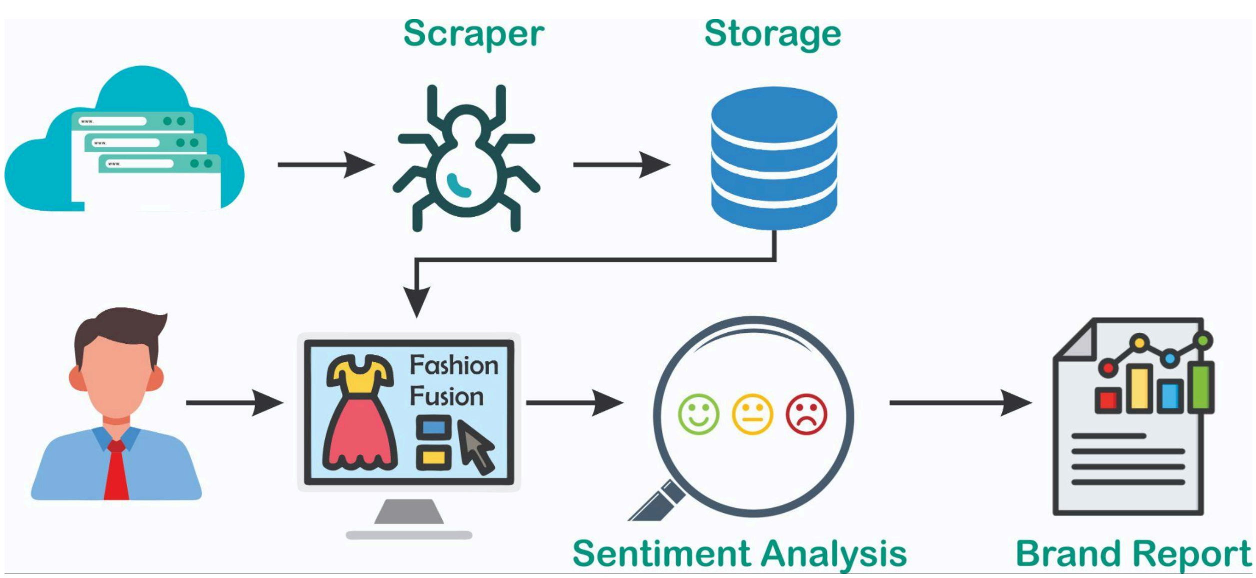
3.1.1 Architectural Style and Module Mapping

The FashionFusion platform follows a **Layered Architecture Pattern** consisting of the following layers:

1. **Presentation Layer**:
   - **Modules**: User Management Module, Chatbot Module, Report Generation Module.
   - **Purpose**: Handles all user interactions and presents data to the user.
2. **Business Logic Layer**:
   - **Modules**: Product Management Module, Review Management Module, Analytics Module.
   - **Purpose**: Contains the core functionality and business rules of the platform.
3. **Data Access Layer**:
   - **Modules**: Product Database, User Reviews Database.
   - **Purpose**: Manages storage and retrieval of data, supporting the business logic.
4. **External Interface Layer**:
   - **Module**: Product Scraper.
   - **Purpose**: Interacts with external sources, such as brand websites, to scrape product data.

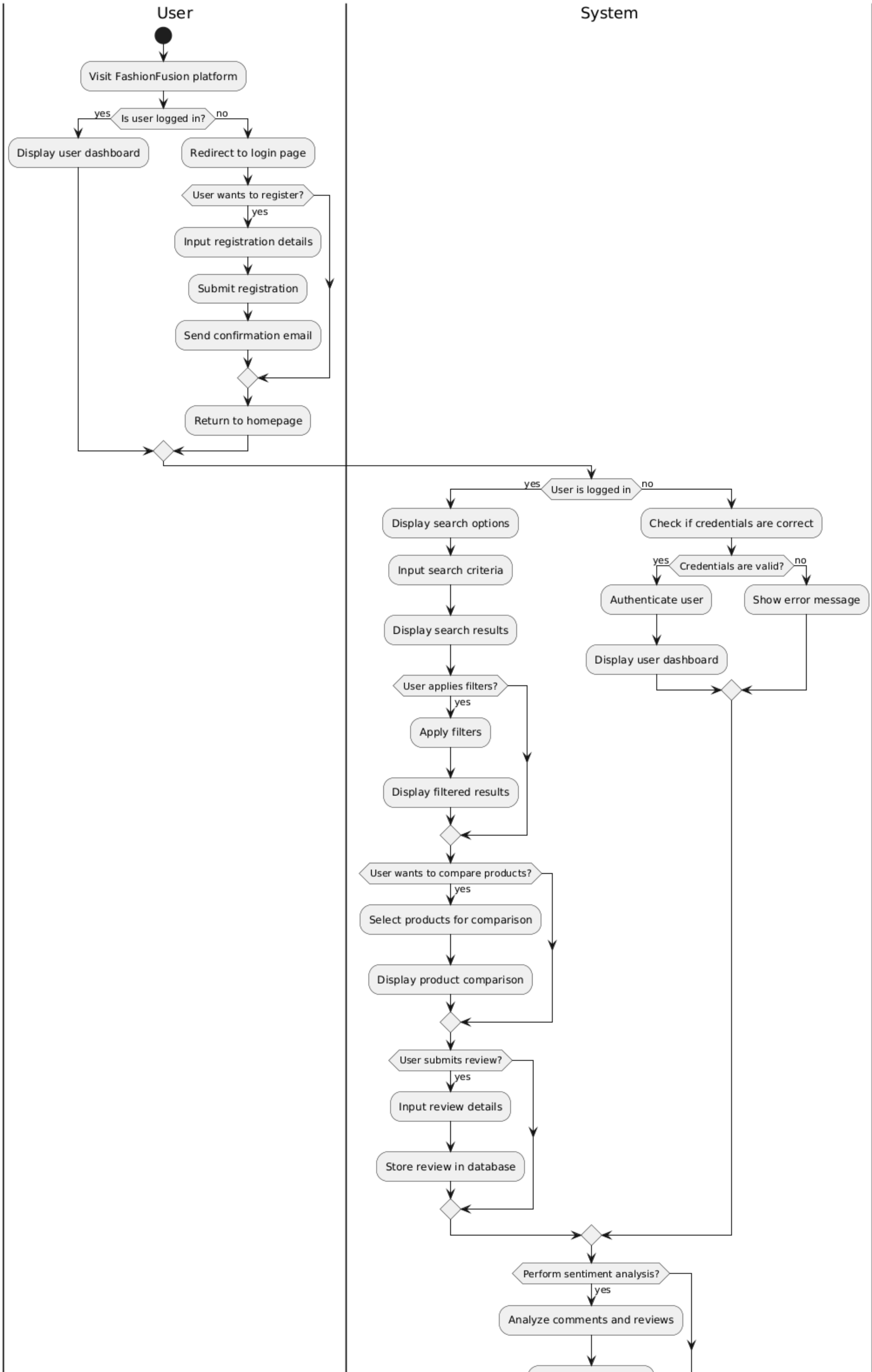
Each module is mapped to a specific layer based on its role and responsibility within the system, facilitating a clear separation of concerns and ease of module management.

Diagram of Architectural Design



3.2 Design Models

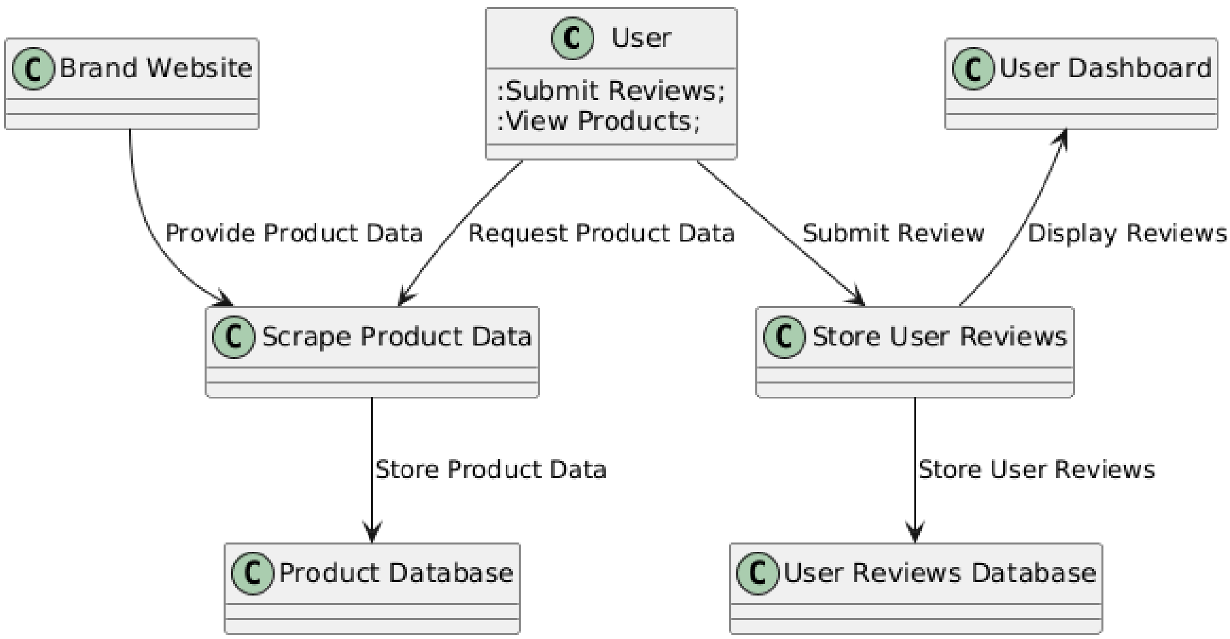
3.2.1 Activity Diagram


User
System
Visit FashionFusion platform
Is user logged in?
yes
no
Display user dashboard
Redirect to login page
User wants to register?
yes
Input registration details
Submit registration
Send confirmation email
Return to homepage
User is logged in
yes
no
Display search options
Input search criteria
Display search results
User applies filters?
yes
Apply filters
Display filtered results
User wants to compare products?
yes
Select products for comparison
Display product comparison
User submits review?
yes
Input review details
Store review in database
Check if credentials are correct
Credentials are valid?
yes
no
Authenticate user
Show error message
Display user dashboard
Perform sentiment analysis?
yes
Analyze comments and reviews

3.2.2 Data Flow Diagram (DFD)

Level 0 Data Flow Diagram

Diagram:



Components

Processes:

- **Process 1: Scrape Product Data**
  Description: This process involves collecting product information from various brand websites and storing it in the database.
- **Process 2: Store User Reviews**
  Description: This process captures user feedback and stores it in the user reviews database.

Sources/Sinks:

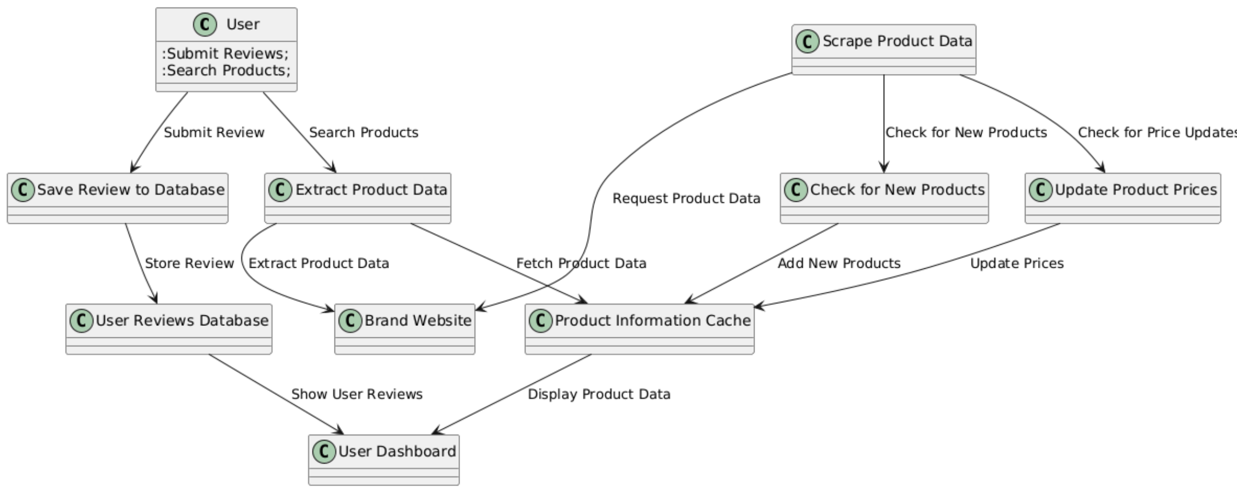
- **Source: Brand Website**
  Description: The external websites from which product data is scraped.
- **Sink: User Dashboard**
  Description: The interface where users can view products, reviews, and their profile information.

Data Stores:

- **Data Store 1: Product Database**
  Description: A repository for all product information scraped from various websites.
- **Data Store 2: User Reviews Database**
  Description: A storage location for user-generated reviews and feedback.
  - ▪

Level 1 Data Flow Diagram

Diagram:



Components

Processes

- **Process 1: Scrape Product Data**
  - **Description:** This process encompasses activities related to retrieving product data from brand websites, including checking for new products and updating existing product prices.
- **Process 2: Check for New Products**
  - **Description:** This subprocess checks the brand websites for any newly added products and stores them in the product information cache.
- **Process 3: Update Product Prices**
  - **Description:** This subprocess verifies whether the prices of existing products have changed and updates the product information cache accordingly.
- **Process 4: Extract Product Data**
  - **Description:** This process extracts detailed product information based on user search queries and displays it in the user dashboard.
- **Process 5: Save Review to Database**

- **Description:** This process captures user-submitted reviews and stores them in the User Reviews Database.

Sources/Sinks

- **Source: Brand Website**
  - **Description:** The external websites from which product data is scraped.
- **Source: User**
  - **Description:** Individuals interacting with the system to submit reviews or search for products.
- **Sink: User Dashboard**
  - **Description:** The interface where users can view their submitted reviews and search results.

Data Stores

- **Data Store 1: Product Information Cache**
  - **Description:** Temporarily holds scraped product data before it is stored permanently in the main product database.
- **Data Store 2: User Reviews Database**
  - **Description:** Stores user-generated reviews and feedback, allowing for easy retrieval and display in the user dashboard.

3.2.3 System Sequence Diagram

Description:
The System-Level Sequence Diagram illustrates the interactions between the user, platform components, and external entities within the FashionFusion system. This diagram highlights how data flows between different system elements during typical operations, such as user interaction with the platform, submission of reviews, and automated product data scraping.

Components:

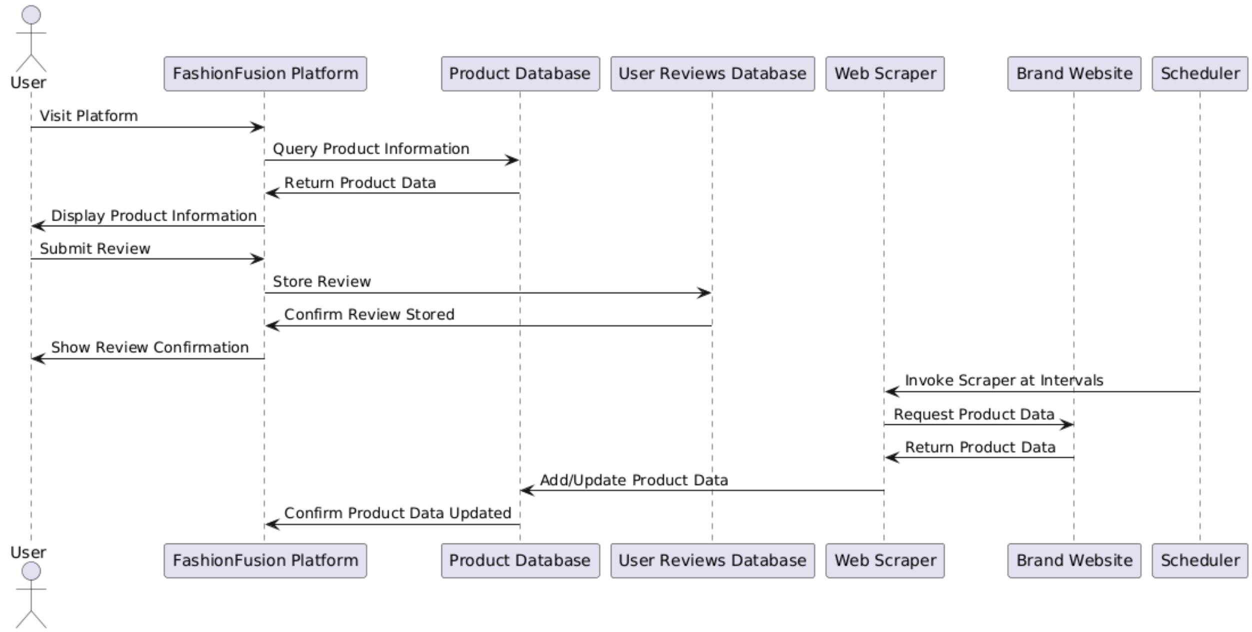
1. User:
   - Represents any individual interacting with the FashionFusion platform, searching for products, and submitting reviews.
2. FashionFusion Platform:
   - Acts as the central hub that manages user interactions, retrieves product data, stores reviews, and communicates with databases.
3. Product Database:
   - Stores detailed product information, including product attributes, prices, and availability. Updated periodically through the web scraper.
4. User Reviews Database:
   - A repository for user-generated reviews, storing feedback and ratings for each product.

5. Web Scraper:
    - Automatically scrapes product data from brand websites. It is invoked by the scheduler at set intervals to update the platform with new products or updated prices.
6. Brand Website:
    - External source where the web scraper gathers product data, including new additions and updated information.
7. Scheduler:
    - Triggers the web scraper at predetermined intervals, ensuring that the product information on the platform remains up-to-date.

Sequence of Interactions:

1. User Interaction with Platform:
    - The user visits the FashionFusion platform to search for products. The platform queries the Product Database and displays the requested product data to the user.
    - When a user submits a review for a product, the platform stores the review in the User Reviews Database and confirms the submission to the user.
2. Automated Web Scraper Invocation:
    - The `Scheduler` component automatically triggers the `Web Scraper` at predefined intervals, without any user interaction.
    - The web scraper requests product data from various `Brand Websites`, which return the relevant product information.
    - This data is then added or updated in the `Product Database`, ensuring that the platform always displays the latest product information to users.

Diagram:



3.3 Data Design

In this section, we will explain how the information domain of the FashionFusion platform is transformed into data structures, detailing how the major data entities are stored, processed, and organized.

Information Domain Transformation

The information domain of the FashionFusion system consists of various entities that represent users, products, reviews, and system-generated insights. These entities are organized into structured data formats, such as tables in a relational database, to facilitate efficient storage, retrieval, and processing.

Major Data Entities

1. **User Entity**:
   - **Attributes**: User ID, Name, Email, Password, Profile Information, etc.
   - **Storage**: Stored in the `Users` table of the database.
2. **Product Entity**:
   - **Attributes**: Product ID, Name, Description, Price, Availability, Brand, etc.
   - **Storage**: Stored in the `Products` table, updated regularly through the scraping processes.
3. **Review Entity**:
   - **Attributes**: Review ID, User ID, Product ID, Rating (1-5), Review Text, Timestamp, etc.
   - **Storage**: Stored in the `Reviews` table, allowing users to submit and manage their reviews.

4. **Analytics Data**:
   - **Attributes**: Metric ID, Product ID, Sentiment Score, Review Count, etc.
   - **Storage**: Generated and stored in an `Analytics` table for reporting purposes.
5. **Chatbot Interaction Logs**:
   - **Attributes**: Log ID, User ID, Query, Response, Timestamp, etc.
   - **Storage**: Stored in a `ChatbotLogs` table to analyze user interactions and improve the recommendation system.

Data Storage

The FashionFusion platform utilizes a relational database management system (RDBMS) for data storage. The major data storage items include:

- **User Database**: Holds user credentials and profile information.
- **Product Database**: Contains detailed information about products scraped from various websites.
- **Review Database**: Stores user-generated reviews linked to specific products.
- **Analytics Database**: Stores computed metrics and insights from the reviews and product data.

Entity-Relationship Diagram (ERD)